RESEARCH PAPER

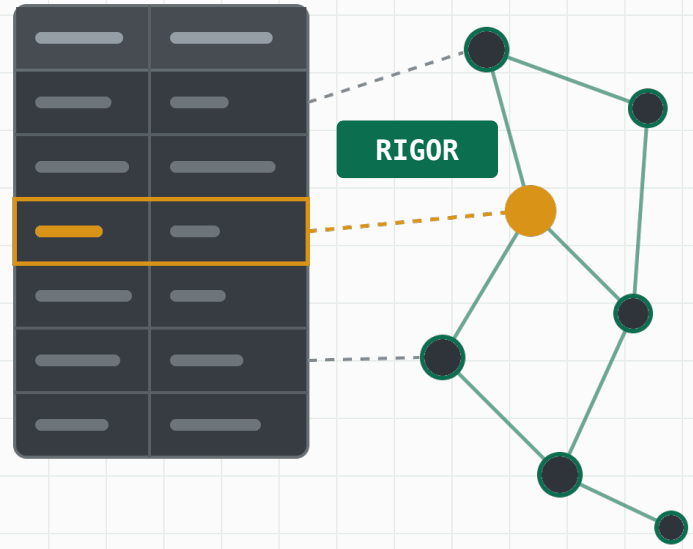
RIGOR

Retrieval-Augmented Generation of Ontologies from Relational Databases

Nadeen Fathallah[1] · Mojtaba Nayyeri[1] · Athish A Yogi[1]
Ratan Bahadur Thapa[1] · Hans-Michael Tautenhahn[3]
Anton Schnurpel[3] · Steffen Staab[1,2]

[1] Institute for Artificial Intelligence, University of Stuttgart, Germany
[2] University of Southampton, United Kingdom
[3] Universitätsklinikum Leipzig, Germany

PAPER arXiv:2506.01232 CODE github.com/NadeenAhmad/RIGOR

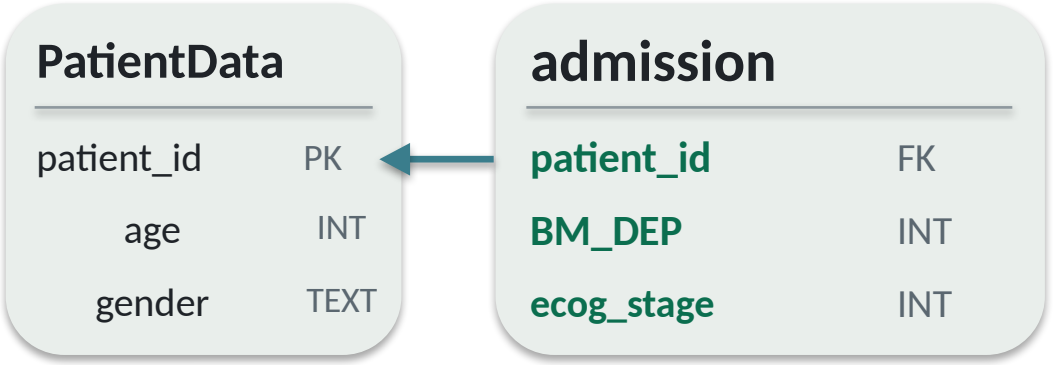


ISWC 2026
Bari, Italy

MOTIVATION / ILLUSTRATIVE CLINICAL EXAMPLE

A database can tell us how data is stored. Can it tell us what the data means?

RELATIONAL SCHEMA · EXCERPT



Illustrative fields. The FK links an admission to a patient.

WHAT WE STILL NEED TO UNDERSTAND

01 **patient_id** — How do we represent the link from an admission to its patient?

02 **BM_DEP** — Which clinical concept does this abbreviation mean?

03 **ecog_stage** — What do the encoded score values mean?

Those meanings matter for ontology-based access, knowledge graphs and integration.

WHAT WE WANT TO BUILD

Not just a cleaner schema. A usable semantic model.

INPUT

admission

patient_id	FK
BM_DEP	INT
ecog_stage	INT

WHAT THIS ENABLES

Concept-based access
Knowledge graph population
Cross-system interoperability

TARGET

Admission —hasPatientData→ PatientData

```
Class: Admission
  SubClassOf: hasPatientData some PatientData        ← schema (foreign key)
  DisjointWith: PatientData                          ← LLM knowledge

DataProperty: BM_DEP
  Annotations:
    rdfs:label "Bone Marrow Depression"@en,          ← documentation
    skos:relatedMatch obo:DOID_4961,                 ← Disease Ontology
    prov:wasDerivedFrom admission:BM_DEP             ← source column

DataProperty: ecog_stage
  Annotations:
    rdfs:label "ECOG Performance Status"@en,         ← documentation
    rdfs:comment "0 = fully active … 5 = dead"@en,   ← LLM knowledge
    prov:wasDerivedFrom admission:ecog_stage         ← source column
```

THE RESEARCH GAP

Two partial solutions. Neither solves the whole problem.

SCHEMA-FIRST RULES

Preserve the structure

Tables, columns and foreign keys can be mapped systematically.

But implicit domain semantics remain missing.

W3C Direct Mapping [Arenas et al. 2012; Sequeda et al. 2012] ·
BootOX [Jiménez-Ruiz et al., ISWC 2015] · **Ben Mahria et al.** [J. Big Data 2021] ·
RODI benchmark [Pinkel et al., SWJ 2017]

LLM-FIRST GENERATION

Suggest the meaning

Labels, concepts and plausible relationships can be proposed.

But elements may be omitted or modeled incorrectly.

LLMs4OL [Babaei Giglou et al., ISWC 2023] · **Mateiu & Groza** [SYNASC 2023] ·
OntoGenix [Val-Calvo et al., IP&M 2025] · **LLM4VKG** [Xiao et al., IJCAI 2025]

OUR QUESTION

Can we preserve schema fidelity and add domain semantics, across the full database, without expert intervention during generation?

OUR ANSWER / RIGOR

Build the ontology one table at a time.

We will construct the answer to our admission example, together.

WHAT WE HAVE

admission

patient_id	FK
BM_DEP	INT
ecog_stage	INT



WHAT WE WANT



+ meaningful labels + encoded values

+ provenance + justified axioms / links

We need to preserve structure, discover meaning, and prevent mistakes, at database scale.

1 ORDER | 2 SEED | 3 RETRIEVE | 4 GENERATE | 5 VALIDATE | 6 MERGE

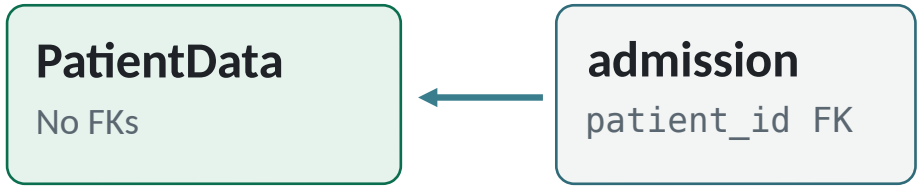
RESULT PREVIEW Ranked first in 70% of expert rankings · highest BURR F1 among compared ontology-learning systems

STEP 1 / ORDER

To understand this table, build its parent first.

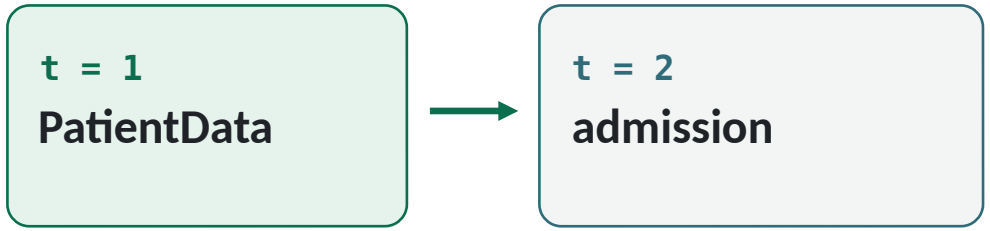
We follow foreign-key dependencies, so known concepts are available when needed.

DATABASE DEPENDENCIES



FK points to parent

PROCESSING ORDER / OUTPUT



Before generating Admission, PatientData already exists in the core ontology.

WHY IT MATTERS The next fragment can reuse known classes and IRIs instead of inventing duplicates.

STEP 2 / SEED

Guarantee schema coverage before LLM enrichment.

Direct Mapping (DM_r) represents every column in the cleaned table before generation.

INPUT / admission (excerpt)

patient_id	FK → PatientData
BM_DEP	INT
ecog_stage	INT

Other columns are also mapped.

OUTPUT / DMr (excerpt) · NO LLM

```
Class: Admission
  SubClassOf:
    patient_id some PatientData,   ← each has a patient
    BM_DEP only xsd:integer,       ← if present, an integer
    ecog_stage only xsd:integer
ObjectProperty: patient_id         ← the FK becomes a link
DataProperty: BM_DEP
DataProperty: ecog_stage
```

GUARANTEE Every cleaned-schema column is represented in the deterministic seed.

STEP 3 / RETRIEVE

The draft is structurally faithful. But what does it mean?

RIGOR gathers a small context package from three complementary sources.

1 DOCUMENTATION

What do column names mean?

BM_DEP → Bone Marrow Depression

Defines cryptic names

2 EXTERNAL ONTOLOGIES

Which domain concepts exist?

ICD-10 · Disease Ontology
Cell and nutrition ontologies

Suggests established concepts

3 GROWING CORE

What have we already defined?

PatientData + previously accepted ontology terms

Reuses established concepts

ONE RETRIEVAL QUERY PER TABLE table + columns + FK targets → top-3 per source

STEP 4 / GENERATE

The LLM enriches a schema-grounded draft.

One prompt combines the direct mapping, retrieved context, and enrichment instructions.

BEFORE / DM_r

```
Class: Admission
  SubClassOf:
    patient_id some PatientData,
    BM_DEP only xsd:integer,
    ecog_stage only xsd:integer
```

Structure only.
Meaning remains implicit.

Gen-LLM

AFTER / ENRICHED ΔO_r (illustrative)

```
ObjectProperty: hasPatientData                                   SCHEMA
  Domain: Admission  Range: PatientData
DataProperty: BM_DEP
  Annotations: rdfs:label "Bone Marrow Depression"@en            DOCS
DataProperty: ecog_stage
  Annotations:
    rdfs:comment "ECOG 0 = fully active,                         DOCS
      1 = restricted, 2 = ambulatory,
      3 = limited self-care,
      4 = completely disabled, 5 = dead"@en,
    prov:wasDerivedFrom src:ecog_stage                           SCHEMA
```

Source tags show context used; provenance traces the source column. External alignments are conditional.

Why validate? Plausible fragments can still break the ontology.

A fluent LLM rarely gets syntax wrong. The risk is fragments that clash with the ontology built so far.

SEMANTIC ERROR · JUDGE-LLM

Generated link for admission:

```
ObjectProperty: hasPatient  Range: Patient
```

Valid OWL, but the core already has PatientData: now two classes mean the same thing. The Judge-LLM sees the core and asks to reuse it.

STRUCTURAL ERROR · GRAPH CHECKS

Later table lab_result reuses a core property:

```
ObjectProperty: hasPatientData  Domain: LabResult
```

Merged with Domain: Admission, OWL infers that every lab result is an admission (pitfall P19). Graph checks allow one domain per property.

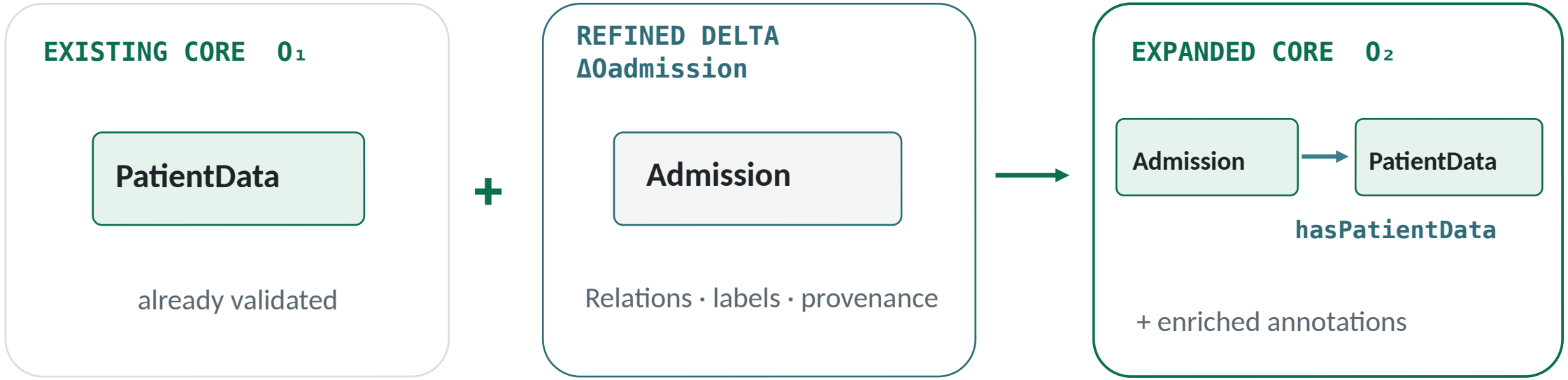
CORRECTION `ObjectProperty: hasPatientData  Range: PatientData` | `ObjectProperty: labResultOf  Domain: LabResult`

Issues → generator correction (up to 2 retries) → graph validation → accepted fragment or fallback

STEP 6 / MERGE

The ontology we built now helps build the next part.

Integrate the refined fragment; use the expanded core as retrieval context.

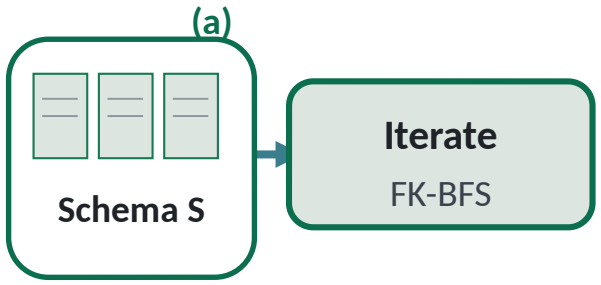


THE LOOP O_2 becomes a retrieval source for the next table → a coherent ontology grows.

1 ORDER | 2 SEED | 3 RETRIEVE | 4 GENERATE | 5 VALIDATE | 6 MERGE

1. Follow the foreign keys

Paper Fig. 1(a) · Start where the foreign-key dependencies begin.



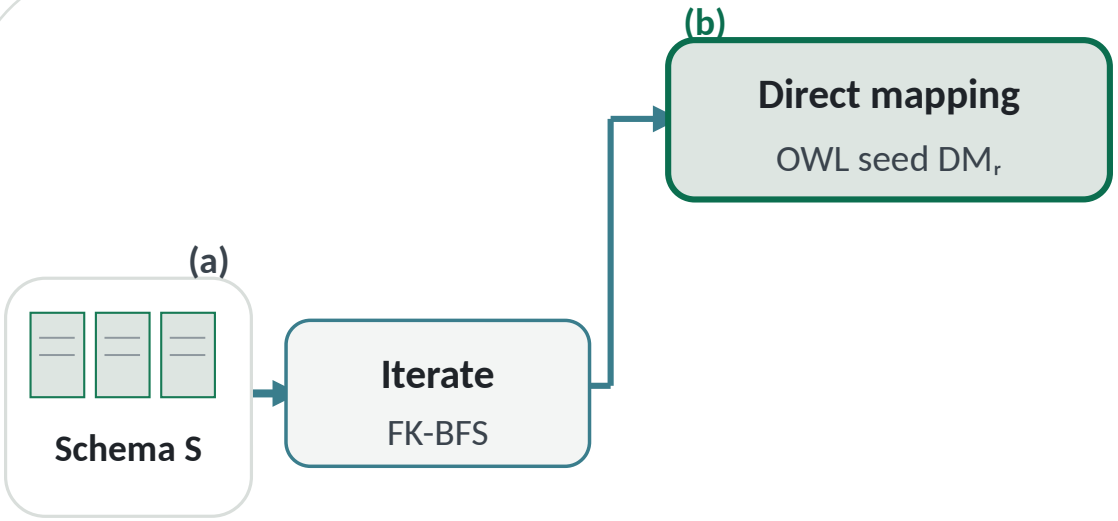
WHY THIS ORDER?

PatientData is processed before admission, so later generation can reuse an existing class.

Recall slide 6 • Parent tables must exist before their dependent tables are processed.

2. Anchor generation in the schema

Paper Fig. 1(b) · The schema-grounded branch feeds the generation step.



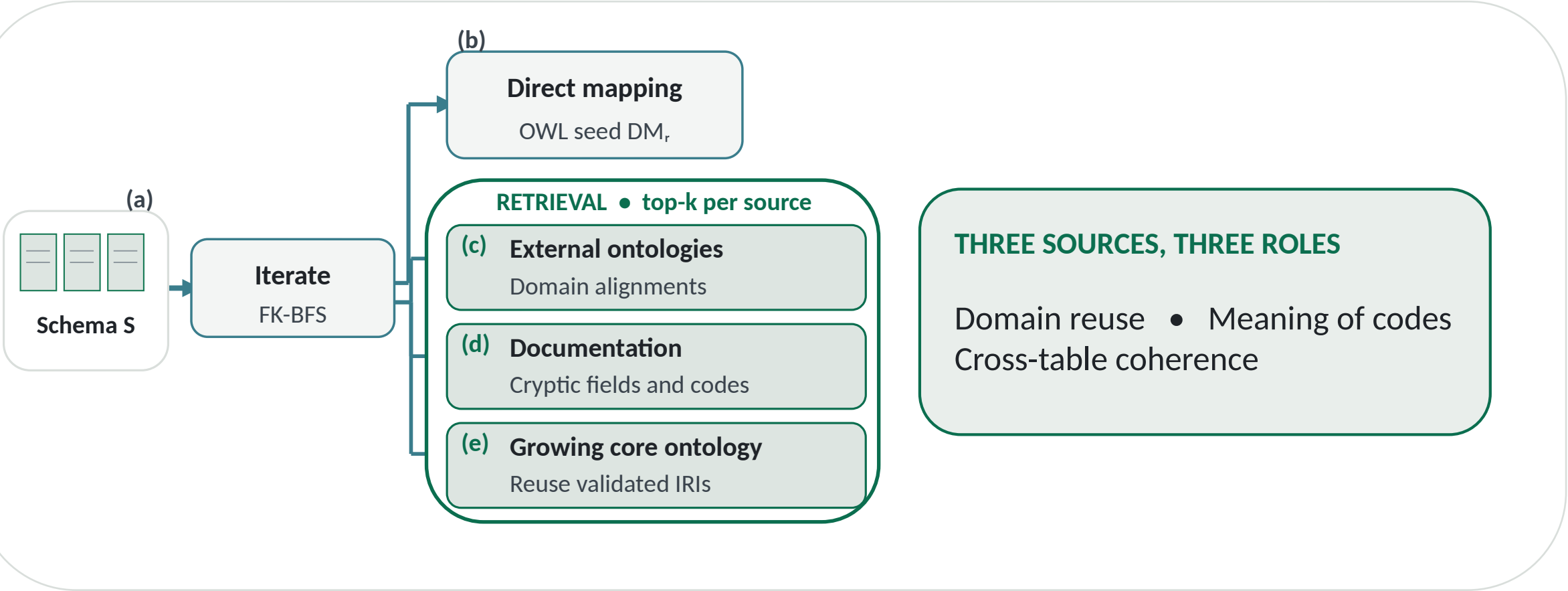
WHY SEED FIRST?

Tables, columns and foreign keys are represented before semantic enrichment.

Recall slide 7 • The deterministic seed covers the cleaned table before LLM enrichment.

3. Retrieve the missing semantics

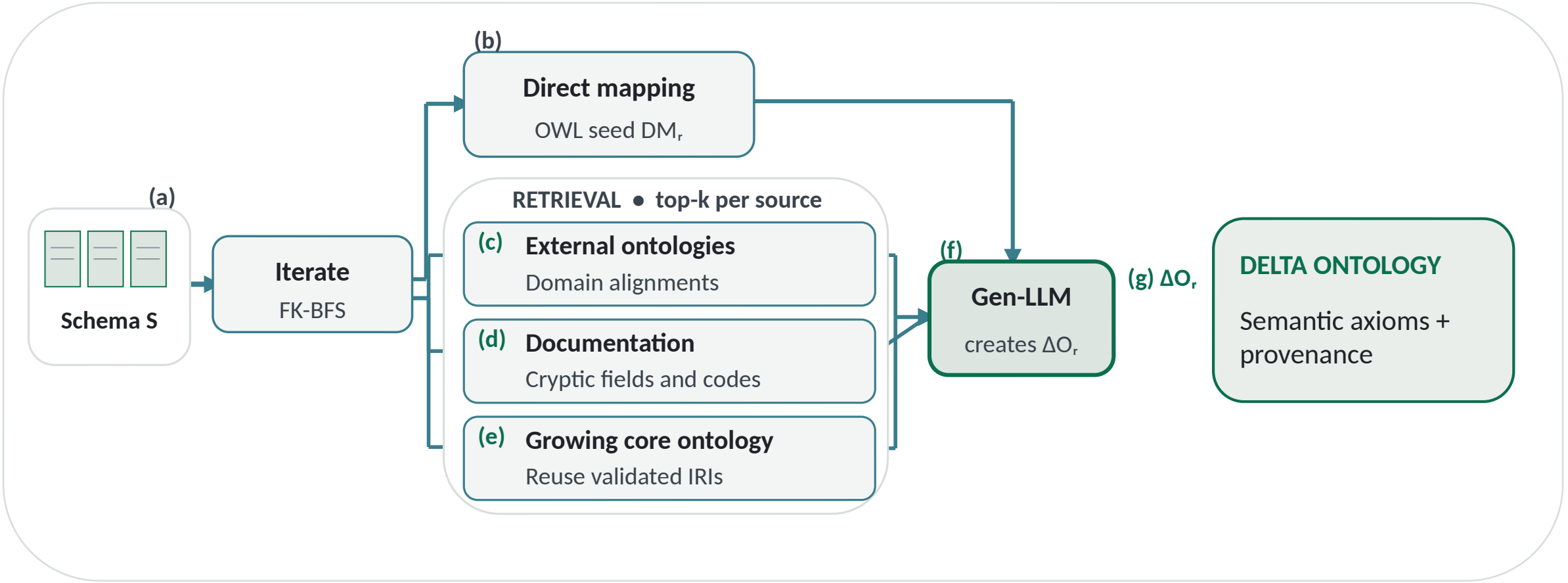
Paper Fig. 1(c–e) · Context is retrieved independently from three sources.



Recall slide 8 • Three sources feed generation: external ontologies, documentation, and core.

4. Generate a delta ontology

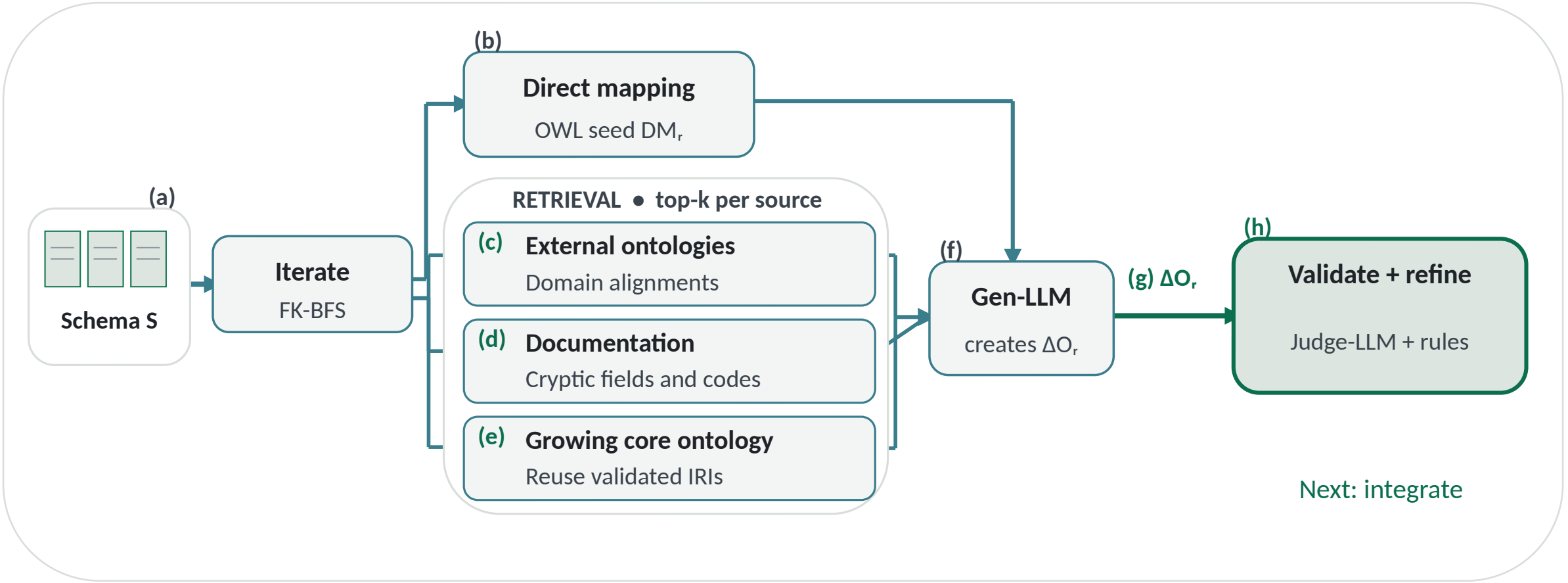
Paper Fig. 1(f–g) · The Gen-LLM receives the seed and retrieved context.



Recall slide 9 • The Gen-LLM enriches the structural seed; the output is a delta ontology.

5. Validate before integration

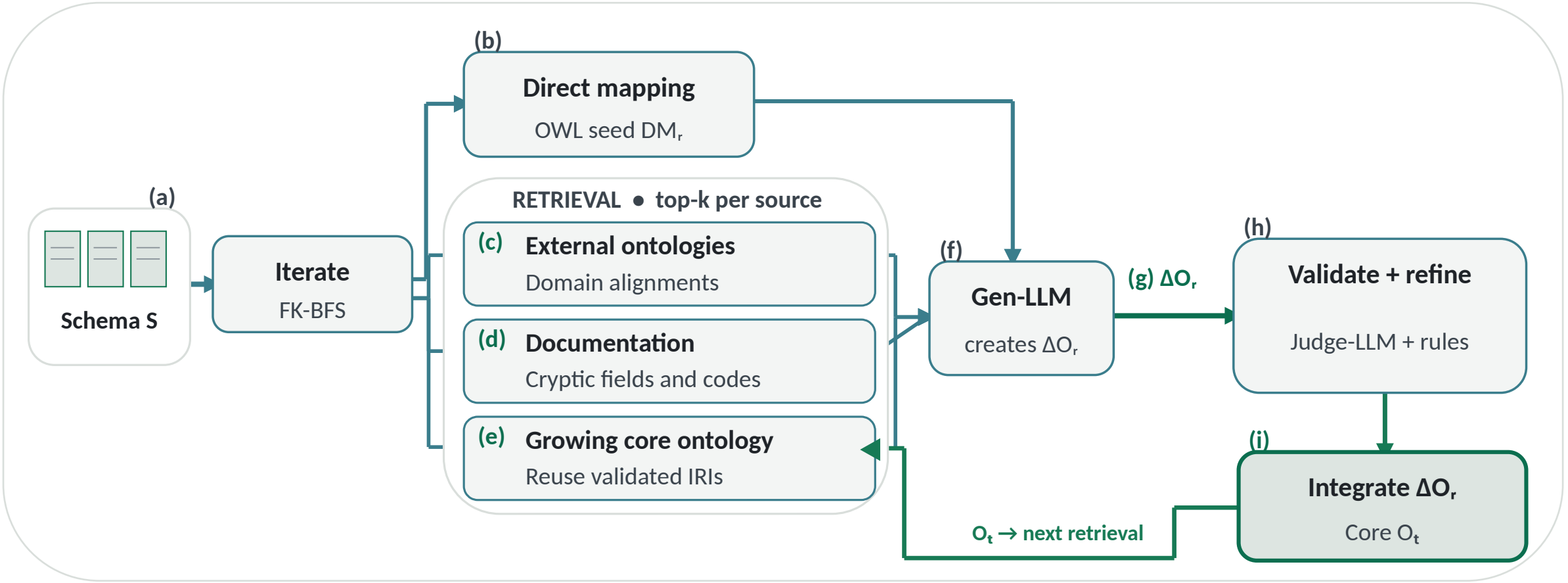
Paper Fig. 1(h) · Validate and refine each delta before it joins the core.



Recall slide 10 • Reject or correct errors before allowing a fragment into the core ontology.

6. Integrate and reuse the growing core

Paper Fig. 1(i) · The growing core feeds the next generation iteration.



Recall slide 11 • The validated result becomes context for generating the next table.

TESTING THE PIPELINE

Does the method work beyond our running example?

Three databases, three generator models, four methods, evaluated from complementary angles.

LIVER CANCER

16 tables · 350 columns

specialist clinical registry

eICU-CRD

31 tables · 559 columns

large ICU schema

CHINOOK

11 tables · 67 columns

music store · second domain

FOUR WAYS TO GENERATE AN ONTOLOGY

1 **Direct Mapping**

Deterministic

→

2 **LLM baseline**

Table only

→

3 **Non-Iterative**

One-pass RAG

→

4 **RIGOR**

Ground + grow + check

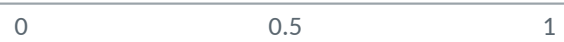
30 ONTOLOGIES quality checks · independent evaluation judge (GPT-5.4) · 10 human experts · BURR benchmark

RESULT 1 / MEANING AND STRUCTURE

Only RIGOR adds meaning without losing structure.

What the generated ontologies contain · 3 databases × 3 LLMs · Direct Mapping: one ontology per database

METHOD	COLUMNS KEPT final schema coverage, range over the runs	LABELS & COMMENTS runs that have them	HIERARCHY + PROVENANCE runs that have them
Direct Mapping	0.78–0.99	○ 0 / 3	○ 0 / 3
LLM baseline	0.31–1.00	○ 0 / 9	○ 0 / 9
Non-Iterative	0.06–0.78	● 9 / 9	○ 0 / 9
RIGOR	**0.77–1.00**	**● 9 / 9**	**● 9 / 9**



MODELING PITFALLS (OOPS!) Fewer than every other method in 9/9 like-for-like runs · zero on Chinook with all three LLMs

Coverage = embedding-based proxy (cosine ≥ 0.55); Direct Mapping is complete by construction. Like-for-like = same database and same LLM.

Sources: paper Tables 7 (pitfalls), 8 (labels, SubClassOf, provenance), 9 (coverage). Raw pitfall counts: backup slide B1.

RESULT 2 / EXTERNAL BENCHMARK

RIGOR keeps the link to the data.

BURR benchmark (Laskowski et al., 2025) · ISWC scenario · highest F1 among the compared ontology-learning systems

WHAT BURR CHECKS

Every ontology element must map back to the database.

CLASSES	`person`	→	`Class: Person`
RELATIONS	`paper.author_id`	→	`ObjectProperty: author`
ATTRIBUTES	`person.name`	→	`DataProperty: name`

Elements match when they map to the same database elements, whatever their names. F1 per element type.

Illustrative elements · ISWC gold mapping: D2RQ example by Bizer & Seaborne

SYSTEM	CLASSES	RELATIONS	ATTRIBUTES
W3C-Mapper / Ontop rule-based	0.53	0.11	0.39
RDB2Onto rule-based	0.67	0.62	0.44
OntoGenix LLM-based	0.58	0.35	0.44
RIGOR LLM + rules	**0.73**	**0.67**	**0.52**

One scenario · 9 tables · 46 columns · 11 FKs · RIGOR with Claude · its prov:wasDerivedFrom links exported as a D2RQ mapping · Source: paper Table 13

RESULT 3 / HUMAN AND LLM EVALUATION

Experts prefer RIGOR's ontologies.

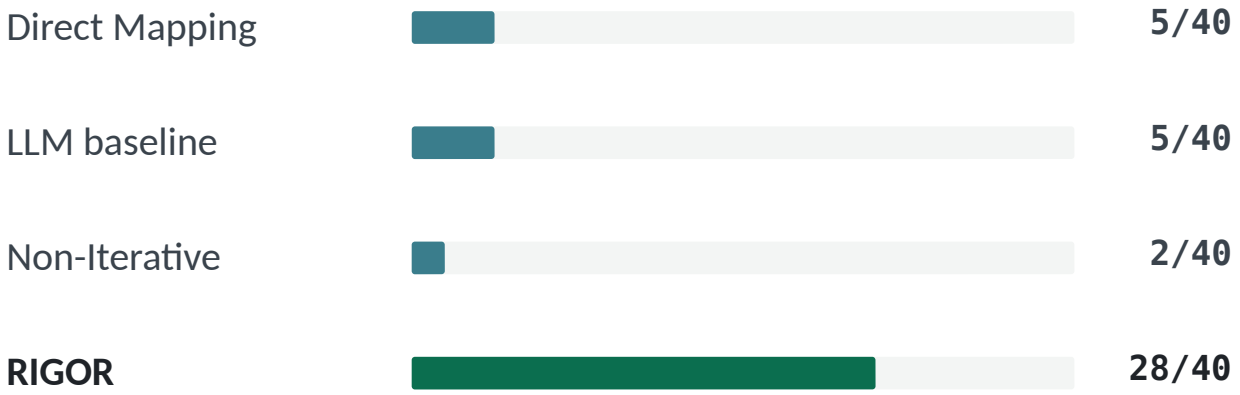
70%

28 of 40

first-place expert rankings

10 experts · 4 medical tables

FIRST PLACE, ACROSS 40 BLIND EXPERT RANKINGS



Expert agreement: Kendall's W = 0.39–0.72

INDEPENDENT LLM JUDGE RIGOR ranked 1st for 58/58 tables · 290 competency questions

FUTURE WORK / FROM SCHEMAS TO INSTANCES

Next: let the data speak.

Today RIGOR reads schemas and documentation. Instance data could propose new knowledge, for experts to review.

1 INSTANCES

What patient rows show

PATIENT	CISPLATIN	BM_DEP AFTER
P01	yes	1
P02	yes	1
P03	no	0
P04	yes	1
P05	no	0

Synthetic illustration. Rows also reveal codes: BM_DEP is only 0 or 1.

2 + DOCUMENTATION

What the drug label says

> *Myelosuppression (bone-marrow suppression) is a documented adverse reaction of cisplatin.*

Cisplatin drug label, paraphrased

Timing is evidence too: BM_DEP is recorded after the first cycle.

3 CANDIDATE KNOWLEDGE

A reviewable hypothesis, not a fact

```
Individual: cisplatin_bmdep
  Types: CandidateAssociation
  Facts:
    hasDrug Cisplatin,
    hasCondition BoneMarrowDepression,
    status "needs expert review"
  Annotations:
    prov:wasDerivedFrom src:BM_DEP,
    prov:wasDerivedFrom doc:cisplatin_label
```

No causes axiom. Co-administered gemcitabine, radiotherapy or cirrhosis can also suppress the marrow.

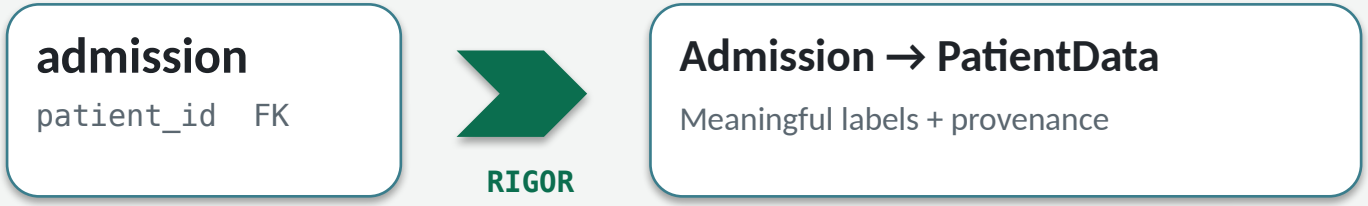
ALSO NEXT Validator and reasoner feedback inside the loop, to catch residual P19 · aggregate statistics only, so patient data stays in the hospital

Instance rows are synthetic. Future directions follow paper Section 5 (instance-level signals, tool feedback, symbolic reasoning).

Thank you.

Questions?

FROM A DATABASE SCHEMA TO AN ONTOLOGY



Meaning without losing structure · link to the data intact · preferred by experts

PAPER + OPEN SOURCE

Paper

GitHub

nadeen.fathallah@ki.uni-stuttgart.de

Raw OOPS! pitfall counts (paper Table 7).

OOPS! findings per ontology (lower is better) · Same dataset and generator model in each comparison

METHOD	LIVER CANCER 16 tables	eICU-CRD 31 tables	CHINOOK 11 tables
Direct Mapping	731	665	102
LLM baseline	422 / 397 / 403	434 / 440 / 421	95 / 74 / 81
Non-Iterative	29 / 61 / 18	39 / 220 / 27	24 / 35 / 14
RIGOR	**20 / 23 / 3**	**2 / 2 / 26**	**0 / 0 / 0**

RIGOR improves on Non-Iterative in 9/9 matched runs; Chinook has zero findings in all 3.

Numbers in each LLM cell: Claude / Mistral / DeepSeek. Direct Mapping uses no LLM.

Interpret carefully: many findings concern missing annotations; critical P19 remains in 3/9 RIGOR runs. Source: Table 7.

BACKUP / B2

Experts score RIGOR highest on average in both databases.

Human expert scores (0–5) per quality dimension · 10 experts, 2 tables per database · best value per row in bold

Liver cancer

DIMENSION	DIRECT MAP.	LLM BASELINE	NON-ITER.	RIGOR
Accuracy	**3.75**	3.30	2.35	**3.75**
Completeness	2.75	3.00	2.25	**3.85**
Conciseness	**4.05**	3.65	2.95	3.60
Adaptability	**2.10**	1.95	1.55	**2.10**
Clarity	2.50	2.45	2.90	**3.90**
Consistency	3.30	2.85	2.35	**3.35**
Domain enrichment	2.20	2.00	2.35	**3.50**
Average	2.95	2.74	2.39	**3.44**

eICU-CRD

DIMENSION	DIRECT MAP.	LLM BASELINE	NON-ITER.	RIGOR
Accuracy	3.05	3.90	0.85	**4.00**
Completeness	2.45	3.10	0.80	**4.10**
Conciseness	3.65	**3.80**	2.05	**3.80**
Adaptability	1.90	2.10	1.00	**2.20**
Clarity	2.55	2.75	1.50	**4.10**
Consistency	2.60	**3.40**	1.30	3.15
Domain enrichment	1.90	2.00	1.15	**3.35**
Average	2.59	3.01	1.24	**3.53**

Where RIGOR is not best: Conciseness (Direct Mapping on liver cancer), Consistency (LLM baseline on eICU-CRD). Source: paper Table 11; LLM-judge scores in Table 10.

BACKUP / B3

Full retrieval gives the best overall alignment, by a small margin.

Retrieval ablation on Chinook with Claude · F1 of alignment between the ontology and each resource (cosine ≥ 0.55)

VARIANT	SCHEMA F1	EXTERNAL F1	DOCS F1	AVG P	AVG R	AVG F1
no retrieval	**1.000**	0.033	0.919	**0.772**	0.622	0.650
only schema + core	0.997	0.033	0.916	0.758	0.622	0.648
only external ontologies	**1.000**	0.033	0.919	0.763	0.622	0.650
only documentation	**1.000**	0.035	**0.921**	0.769	0.624	0.652
full RIGOR	0.993	**0.059**	0.914	0.756	**0.630**	**0.655**

Read with care: schema alignment is saturated because every variant keeps the direct-mapping seed, and external-ontology F1 is low in absolute terms. Full RIGOR is best on external F1, average recall and average F1.

Source: paper Table 14 (Chinook, Claude). One database and one model.

BACKUP / B4

All nine RIGOR runs together cost about 33 dollars.

Generation cost and runtime per database and model, including Gen-LLM and Judge-LLM calls

DATABASE	MODEL	REQUESTS	TOKENS	COST (USD)	RUNTIME (MIN)
eICU-CRD	Claude Opus 4.6	168	1,660,000	$21.30	182.4
eICU-CRD	Mistral-Small-24B	96	1,290,000	$0.08	188.4
eICU-CRD	DeepSeek-V3	52	334,000	$0.18	20.6
Liver cancer	Claude Opus 4.6	70	731,000	$10.30	94.4
Liver cancer	Mistral-Small-24B	34	246,000	$0.02	20.5
Liver cancer	DeepSeek-V3	33	142,000	$0.07	23.8
Chinook	Claude Opus 4.6	24	119,000	$1.28	9.0
Chinook	Mistral-Small-24B	53	344,000	$0.02	65.0
Chinook	DeepSeek-V3	22	71,000	$0.03	12.0
Total		**552**	**4,937,000**	**$33.28**	**616.1**

Source: paper Table 15. Totals are summed from the nine rows; the paper's printed totals (651 requests, $34.28) also include runs not listed in the table.

Hard questions (1/2)

Short, honest answers · full versions in presentation/SPEECH.md

The Judge-LLM is the same model as the generator. Isn't that self-evaluation?

Inside the loop, yes: the same model generates and judges, so RIGOR needs no second model. Two things do not depend on it. The rule-based graph checks are deterministic. And the evaluation used a different model family, GPT-5.4, plus ten human experts, who ranked RIGOR first in 70% of cases.

Is the Non-Iterative baseline a strawman?

It receives the same retrieved context, but its prompt asks for less (labels and comments). Its biggest failure is dropping columns, down to 6%, which RIGOR prevents by construction with the seed. A prompt-matched ablation is a fair next experiment.

Could the LLM judge have position bias?

Fragments were anonymized, but the order was fixed (paper Fig. 7), so we do not rely on the LLM judge alone. The human study points the same way: 28 of 40 first places, with significant agreement. Future runs should randomize the order.

Why do critical P19 pitfalls remain (multiple domains or ranges)?

They appear in 3 of 9 runs. A likely source is the union merge: when a property such as a patient link is reused across tables, each table adds its own domain. Merge-time checks and validator feedback inside the loop are the planned fix.

Why only schemas and documentation, not the data?

Portability and privacy. Clinical data often cannot leave the hospital, while schemas and documentation can. Instance-level signals, computed as aggregates, are our next step.

Hard questions (2/2)

Short, honest answers · full versions in presentation/SPEECH.md

Column coverage drops to 0.77. Why?

That is DeepSeek on eICU. The seed is complete by construction, but enrichment can drop columns, and coverage is an embedding-based proxy. Stronger per-fragment coverage checks are planned.

Could the LLMs have seen these databases in training?

eICU and Chinook are public. The liver-cancer registry is private, so no model saw it, and the LLM judge still ranked RIGOR first on all 16 of its tables.

Are the external ontologies actually used?

Retrieved external concepts are passed as context, and alignment axioms are added only on a good match. In the ablation, full RIGOR has the highest external-ontology alignment (F1 0.059 vs 0.033 without retrieval), though absolute values are small.

How does the traversal handle self-referencing foreign keys (Chinook Employee.ReportsTo)?

A self-reference does not wait on another table, so it does not affect the order; it becomes an object property from the class to itself, which the prompt allows for meaningful recursive relations.

Why not compare with Gemini 2.5 Pro on BURR?

In BURR it is a general-purpose LLM prompted to write the mapping directly: a mapping generator, not an ontology-learning system. Against it, RIGOR's class F1 is higher (.73 vs .65), relations are on par (.67 vs .68), attributes lower (.52 vs .56), and RIGOR also produces labels, hierarchy and provenance.